

FinAgent AI Analysis Report

Generated	2026-05-23
Tickers	AAPL, MSFT, NVDA, GOOG
Macro	GC=F, CL=F
Period	5y

1. Market Trend Summary

The market trend summary highlights a bullish direction over the past 5 years, with all stocks exhibiting positive returns. The overall momentum is driven by NVDA, which leads the pack with a remarkable 1006.84% return, fueled by its high mean daily return of 5.15%. In contrast, MSFT lags behind with a 53.08% return, largely due to its relatively lower mean daily return of 1.04%.

From a technical perspective, the Relative Strength Index (RSI) signals overbought conditions for AAPL (71.5), NVDA (71.7), and GOOG (75.0), which may indicate potential pullbacks. Conversely, MSFT's RSI (50.1) suggests a neutral stance.

Volatility observations reveal a range-bound pattern, with NVDA exhibiting the highest annualized volatility of 36.56%, while MSFT shows the lowest volatility among the group at 21.77%. The absence of outlier days across all stocks suggests a relatively stable market environment. Overall, the market demonstrates a positive trajectory, with some stocks potentially due for correction amid overbought conditions. Investors should monitor RSI levels and volatility to inform their investment decisions.

2. Anomaly Detection

Analysis:

Given the provided anomaly data, there are no flagged anomalous trading days for the specified stocks ['AAPL', 'MSFT', 'NVDA', 'GOOG'] as the note indicates "No outliers detected — data appears clean".

Since no outliers are detected, speculation on probable causes for anomalies is not applicable in this case.

Assessment:

The data quality and reliability appear to be high, as the analysis did not detect any outliers or anomalies that exceed $3 \times \text{IQR}$ or $|\text{return}| > 40\%$. This suggests that the data is consistent and free from significant errors or unusual patterns.

3. Risk Commentary & Asset Comparison

Risk-Adjusted Return Analysis Report

Introduction

This report provides a comprehensive analysis of four assets (AAPL, MSFT, NVDA, and GOOG) from a risk-adjusted return perspective. The analysis is based on data from July 1, 2021, to May 1, 2026.

Risk-Return Ranking

To rank the assets by risk-adjusted return, we calculate the Sharpe Ratio, assuming a 0% risk-free rate. The Sharpe Ratio is calculated as (period return / annualized volatility).

Asset	Period Return (%)	Annualized Volatility	Sharpe Ratio
NVDA	1006.84	0.3656	2.75
GOOG	182.87	0.3107	0.59
AAPL	117.21	0.2028	0.58
MSFT	53.08	0.2177	0.24

Ranking (highest to lowest):

1. NVDA
2. GOOG
3. AAPL
4. MSFT

Diversification Assessment

To assess diversification, we examine the relationship between the assets' returns. While the data does not provide direct correlation coefficients, we can infer from the returns and volatilities that:

- NVDA and GOOG have higher returns and volatilities, suggesting they may be more closely related to each other.
- AAPL and MSFT have lower returns and volatilities, indicating they might be more independent of the other two assets.

A diversified portfolio would likely include a mix of these assets to minimize risk.

Investor Suitability

Based on the analysis:

- **Conservative Investor:** AAPL is the most suitable asset, with a relatively lower volatility (0.2028) and a moderate return (117.21%). This asset provides a more stable return profile.
- **Aggressive Investor:** NVDA is the most suitable asset, with an exceptionally high return (1006.84%) and a higher volatility (0.3656). This asset offers a high potential return, but with increased risk.

Risk Warnings

The following risk warnings are noted:

- **High RSI:** NVDA (71.7), AAPL (71.5), and GOOG (75.0) have high RSI values, indicating potential overbought conditions. This may lead to a correction in the short term.

- **High Volatility:** NVDA (0.3656) and GOOG (0.3107) exhibit higher volatility, making them more susceptible to significant price fluctuations.
- **No significant negative return trends** are observed in the given data.

In conclusion, this analysis provides a comprehensive risk-adjusted return assessment of the four assets. Investors should consider their individual risk tolerance and investment goals when making decisions based on this report.

4. Fundamental Valuation Analysis

VALUATION: Based on the price-to-earnings (P/E) ratio, 'MSFT' appears to be the cheapest with a forward P/E of 21.65, while 'AAPL' is the most expensive with a forward P/E of 32.16. Considering the P/E ratios, 'MSFT' offers the best value, trading at a 34% discount to 'AAPL' and 18% discount to 'GOOG'. 'NVDA' has the lowest forward P/E of 17.03, indicating it offers the most attractive valuation.

PROFITABILITY: The companies' profit margins are: 'NVDA' (62.97%), 'MSFT' (39.34%), 'GOOG' (37.92%), and 'AAPL' (27.15%). 'NVDA' converts revenue to profit most efficiently, with a gross margin of 74.14%, significantly outpacing its peers. 'MSFT' and 'GOOG' also demonstrate strong profitability, while 'AAPL' lags behind.

EARNINGS QUALITY: EPS levels are: 'MSFT' (\$16.78), 'GOOG' (\$13.13), 'AAPL' (\$8.25), and 'NVDA' (\$6.52). 'MSFT' has the highest EPS, indicating strong earnings quality. 'GOOG' and 'AAPL' also exhibit respectable EPS levels, while 'NVDA' has the lowest, potentially due to its high growth investments.

RISK: The Debt/Equity ratios are: 'AAPL' (79.55), 'GOOG' (20.03), 'MSFT' (30.27), and 'NVDA' (6.55). Beta values are: 'NVDA' (2.24), 'GOOG' (1.27), 'MSFT' (1.09), and 'AAPL' (1.06). 'NVDA' has the highest Beta, indicating the highest market risk, while 'AAPL' has the highest financial risk due to its elevated Debt/Equity ratio.

VERDICT: Ranked from most to least attractive, the stocks are:

1. 'NVDA' - Offers the most attractive valuation and exceptional profitability.
2. 'MSFT' - Provides a strong combination of valuation, profitability, and earnings quality.
3. 'GOOG' - Exhibits a solid balance of profitability and earnings quality, albeit at a higher valuation.
4. 'AAPL' - Lags behind due to its rich valuation, lower profitability, and elevated financial risk.

5. Macro Environment Analysis

Given the current macroeconomic landscape, here's a concise analysis:

1. **Monetary Policy:** Although the current Fed Funds Rate is unknown, the 6mo_ago rate of 3.88% suggests a tightening bias. Rising interest rates can lead to decreased equity valuations, particularly for growth stocks.
2. **Inflation:** With the current CPI YoY Inflation unknown but 2.99% six months ago, sectors like Consumer Staples and Healthcare tend to benefit from moderate inflation, while sectors like Technology may suffer due to decreased consumer spending power.
3. **Treasury Yield:** The 10Y Treasury Yield has increased by 0.55% to 4.57%. This rise may lead to P/E compression, impacting growth stocks like NVDA and GOOG. However, without a clear yield curve inversion signal, recession risks are not immediately apparent.

4. **Labour Market:** The Unemployment Rate was 4.5% six months ago, which is relatively low. Although the current rate is unknown, a low unemployment rate generally indicates a strong labour market, reducing recession risk.

5. **Portfolio Impact:** NVDA and GOOG are most exposed to macro risk due to their growth profiles and sensitivity to interest rate changes. In contrast, MSFT and AAPL may be less exposed, given their diversified business models and relatively stable cash flows.

In conclusion, the **MACRO RISK RATING** is **Medium**, driven by potential P/E compression from rising Treasury yields and unclear monetary policy and inflation trends, which may impact the portfolio's growth-oriented stocks.

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